

Trigonometry



The unit circle and radian measure

- 1 Convert 210° to radians.
 - 2 Convert $\frac{5\pi}{4}$ radians to degrees.
 - 3 Find the exact value of $\sin(240^\circ)$.
-

Trigonometric identities

- 4 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$.
 - 5 Simplify $\tan \theta \cdot \cos \theta$.
 - 6 If $\sin \theta = \frac{3}{5}$ and θ is in Quadrant II, find $\cos \theta$ and $\tan \theta$.
-

Double angle formulas

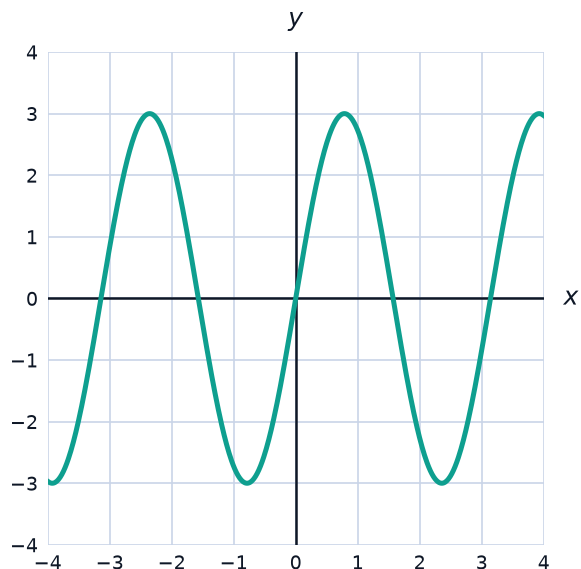
- 7 Given $\cos \theta = \frac{5}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.
 - 8 Use double-angle formulas to express $\sin 4\theta$ in terms of $\sin \theta$ and $\cos \theta$ only.
-

Solving trigonometric equations

- 9 Solve $2\sin \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
 - 10 Solve $2\cos^2 \theta + \cos \theta - 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.
-

Graphs of trigonometric functions

- 11 The graph shows $y = 3\sin(2x)$. State the amplitude and period.



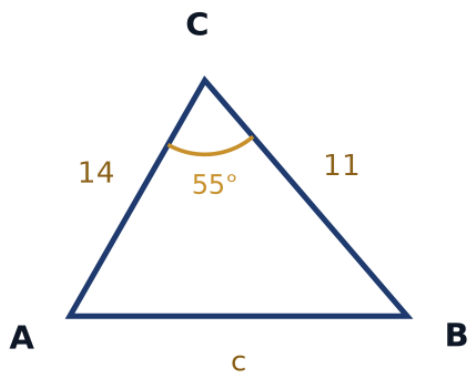
- 12 The function $y = 4\cos(3x - \pi)$ has what amplitude, period, and phase shift?

The Law of Sines

- 13 In triangle ABC , $a = 10$, $A = 40^\circ$, $B = 65^\circ$. Find b , to the nearest hundredth.
- 14 In triangle ABC , $a = 8$, $b = 10$, $A = 35^\circ$. Determine how many triangles are possible, and find angle B (to the nearest tenth of a degree) for each.

The Law of Cosines and the sine area formula

- 15 In triangle ABC shown below, $a = 11$, $b = 14$, and $C = 55^\circ$. Find side c , to the nearest hundredth, and the area of the triangle, to the nearest tenth.



- 16 State the two formulas used above (Law of Cosines for a side, and the sine formula for area) in general form for a triangle with sides a , b , c and included angle C .
-

Inverse trigonometric functions

- 17 Evaluate $\arcsin\left(\frac{\sqrt{2}}{2}\right)$ exactly, in radians.
- 18 Evaluate $\arccos(-1)$ exactly, in radians.
- 19 Find $\theta = \arctan(1)$ in degrees (principal value).
-

A well-designed word problem: locating a ship from two lighthouses

- 20 This question has three parts. Two lighthouses A and B are 15 km apart along a straight coastline. A ship at point C is sighted such that $\angle CAB = 72^\circ$ and $\angle CBA = 58^\circ$ (both measured from line AB toward the ship). (a) Find $\angle ACB$. (b) Use the Law of Sines to find the distance AC from lighthouse A to the ship, to the nearest tenth of a km. (c) Find the distance BC from lighthouse B to the ship, to the nearest tenth of a km.
-

Multiple choice — easy

- 21 Convert 135° to radians.

- a. $\frac{3\pi}{5}$
- b. $\frac{3\pi}{2}$
- c. $\frac{3\pi}{4}$
- d. π

- 22 Convert 60° to radians.

- a. $\frac{2\pi}{3}$
- b. π
- c. $\frac{\pi}{4}$
- d. $\frac{\pi}{3}$

23 Convert 345° to radians.

- a. 2π
- b. $\frac{23\pi}{13}$
- c. $\frac{23\pi}{12}$
- d. $\frac{23\pi}{6}$

24 Convert 105° to radians.

- a. $\frac{2\pi}{3}$
- b. $\frac{7\pi}{13}$
- c. $\frac{7\pi}{6}$
- d. $\frac{7\pi}{12}$

25 Convert 90° to radians.

- a. π
- b. $\frac{3\pi}{2}$
- c. $\frac{\pi}{3}$
- d. $\frac{\pi}{2}$

26 Convert 30° to radians.

- a. $\frac{\pi}{3}$
- b. $\frac{\pi}{7}$
- c. $\frac{\pi}{2}$
- d. $\frac{\pi}{6}$

27 Convert 315° to radians.

a. $\frac{7\pi}{5}$

b. $\frac{7\pi}{2}$

c. 2π

d. $\frac{7\pi}{4}$

28 Convert 45° to radians.

a. $\frac{\pi}{4}$

b. $\frac{\pi}{5}$

c. $\frac{3\pi}{4}$

d. $\frac{\pi}{2}$

29 Convert 165° to radians.

a. $\frac{11\pi}{6}$

b. $\frac{11\pi}{12}$

c. π

d. $\frac{11\pi}{13}$

30 Convert 225° to radians.

a. $\frac{5\pi}{2}$

b. $\frac{5\pi}{4}$

c. π

d. $\frac{3\pi}{2}$

31 Convert 150° to radians.

a. $\frac{5\pi}{7}$

b. $\frac{5\pi}{3}$

c. π

d. $\frac{5\pi}{6}$

32 Convert 15° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{13}$

c. $\frac{\pi}{4}$

d. $\frac{\pi}{12}$

33 Convert 75° to radians.

a. $\frac{5\pi}{12}$

b. $\frac{5\pi}{13}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{2}$

34 Convert 210° to radians.

a. π

b. $\frac{4\pi}{3}$

c. $\frac{7\pi}{3}$

d. $\frac{7\pi}{6}$

35 Convert 240° to radians.

a. $\frac{5\pi}{3}$

b. $\frac{4\pi}{3}$

c. π

d. $\frac{8\pi}{3}$

Multiple choice — medium

36 Solve $\tan \theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

a. $120^\circ, 300^\circ$

b. $45^\circ, 225^\circ$

c. $30^\circ, 210^\circ$

d. $135^\circ, 315^\circ$

37 Solve $\tan \theta = -\frac{\sqrt{3}}{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 210^\circ$

b. $150^\circ, 330^\circ$

c. $135^\circ, 315^\circ$

d. $120^\circ, 300^\circ$

38 Solve $\tan \theta = -\sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $150^\circ, 330^\circ$

b. $120^\circ, 300^\circ$

c. $135^\circ, 315^\circ$

d. $60^\circ, 240^\circ$

39 Solve $\sin \theta = -\frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $225^\circ, 315^\circ$

b. $210^\circ, 330^\circ$

c. $30^\circ, 150^\circ$

d. $45^\circ, 135^\circ$

40 Solve $\cos \theta = -\frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 330^\circ$

b. $135^\circ, 225^\circ$

c. $150^\circ, 210^\circ$

d. $120^\circ, 240^\circ$

41 Solve $\cos \theta = \frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $150^\circ, 210^\circ$

c. $45^\circ, 315^\circ$

d. $60^\circ, 300^\circ$

42 Solve $\cos \theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $135^\circ, 225^\circ$

b. $30^\circ, 330^\circ$

c. $120^\circ, 240^\circ$

d. $60^\circ, 300^\circ$

43 Solve $\cos \theta = \frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $150^\circ, 210^\circ$

b. $30^\circ, 330^\circ$

c. $45^\circ, 315^\circ$

d. $120^\circ, 240^\circ$

44 Solve $\sin \theta = \frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

a. $30^\circ, 150^\circ$

b. $240^\circ, 300^\circ$

c. $60^\circ, 120^\circ$

d. $210^\circ, 330^\circ$

45 Solve $\tan \theta = \sqrt{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $60^\circ, 240^\circ$

b. $135^\circ, 315^\circ$

c. $150^\circ, 330^\circ$

d. $120^\circ, 300^\circ$

46 Solve $\tan \theta = \frac{\sqrt{3}}{3}$ for $0^\circ \leq \theta < 360^\circ$.

a. $120^\circ, 300^\circ$

b. $30^\circ, 210^\circ$

c. $135^\circ, 315^\circ$

d. $60^\circ, 240^\circ$

47 Solve $\tan \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

- a. $60^\circ, 240^\circ$
- b. $120^\circ, 300^\circ$
- c. $45^\circ, 225^\circ$
- d. $150^\circ, 330^\circ$

48 Solve $\sin \theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $210^\circ, 330^\circ$
- b. $240^\circ, 300^\circ$
- c. $225^\circ, 315^\circ$
- d. $60^\circ, 120^\circ$

49 Solve $\sin \theta = \frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $60^\circ, 120^\circ$
- b. $30^\circ, 150^\circ$
- c. $45^\circ, 135^\circ$
- d. $240^\circ, 300^\circ$

50 Solve $\sin \theta = -\frac{\sqrt{2}}{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $210^\circ, 330^\circ$
- b. $45^\circ, 135^\circ$
- c. $60^\circ, 120^\circ$
- d. $225^\circ, 315^\circ$

Multiple choice — hard

- 51 In triangle ABC , $a = 18$, $A = 80^\circ$, $B = 70^\circ$. Find b , to the nearest hundredth.
- a. 17.68
 - b. 18.18
 - c. 17.18
 - d. 16.68
- 52 In triangle ABC , $a = 17$, $A = 20^\circ$, $B = 80^\circ$. Find b , to the nearest hundredth.
- a. 48.95
 - b. 49.95
 - c. 49.45
 - d. 48.45
- 53 In triangle ABC , $a = 15$, $A = 60^\circ$, $B = 30^\circ$. Find b , to the nearest hundredth.
- a. 9.16
 - b. 8.66
 - c. 8.16
 - d. 9.66
- 54 In triangle ABC , $a = 11$, $A = 55^\circ$, $B = 60^\circ$. Find b , to the nearest hundredth.
- a. 11.13
 - b. 12.13
 - c. 12.63
 - d. 11.63

- 55 In triangle ABC , $a = 17$, $A = 80^\circ$, $B = 45^\circ$. Find b , to the nearest hundredth.
- a. 12.71
 - b. 12.21
 - c. 13.21
 - d. 11.71
- 56 In triangle ABC , $a = 9$, $b = 15$, and $C = 110^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.
- a. 18.96, 61.4
 - b. 19.96, 65.4
 - c. 20.96, 63.4
 - d. 19.96, 63.4
- 57 In triangle ABC , $a = 13$, $b = 11$, and $C = 130^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.
- a. 21.77, 56.8
 - b. 21.77, 54.8
 - c. 20.77, 52.8
 - d. 22.77, 54.8
- 58 In triangle ABC , $a = 13$, $b = 9$, and $C = 80^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.
- a. 14.47, 57.6
 - b. 15.47, 57.6
 - c. 14.47, 59.6
 - d. 13.47, 55.6

- 59 In triangle ABC , $a = 14$, $b = 10$, and $C = 60^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.
- 12.49, 60.6
 - 13.49, 60.6
 - 12.49, 62.6
 - 11.49, 58.6
- 60 In triangle ABC , $a = 9$, $b = 12$, and $C = 70^\circ$. Find side c , to the nearest hundredth, and the area, to the nearest tenth.
- 12.29, 52.7
 - 11.29, 48.7
 - 13.29, 50.7
 - 12.29, 50.7
- 61 Given $\cos \theta = \frac{35}{37}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.
- $\frac{840}{1369}, \frac{1081}{1369}$
 - $\frac{840}{1369}, -\frac{1081}{1369}$
 - $-\frac{840}{1369}, \frac{1081}{1369}$
 - $\frac{1081}{1369}, \frac{840}{1369}$
- 62 Given $\cos \theta = \frac{24}{25}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.
- $-\frac{336}{625}, \frac{527}{625}$
 - $\frac{336}{625}, \frac{527}{625}$
 - $\frac{336}{625}, -\frac{527}{625}$
 - $\frac{527}{625}, \frac{336}{625}$

63 Given $\cos \theta = \frac{15}{17}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $\frac{240}{289}, -\frac{161}{289}$

b. $-\frac{240}{289}, \frac{161}{289}$

c. $\frac{240}{289}, \frac{161}{289}$

d. $\frac{161}{289}, \frac{240}{289}$

64 Given $\cos \theta = \frac{12}{13}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{120}{169}, \frac{119}{169}$

b. $\frac{120}{169}, -\frac{119}{169}$

c. $\frac{119}{169}, \frac{120}{169}$

d. $\frac{120}{169}, \frac{119}{169}$

65 Given $\cos \theta = \frac{4}{5}$ with θ in Quadrant I, find $\sin 2\theta$ and $\cos 2\theta$.

a. $-\frac{24}{25}, \frac{7}{25}$

b. $\frac{24}{25}, \frac{7}{25}$

c. $\frac{24}{25}, -\frac{7}{25}$

d. $\frac{7}{25}, \frac{24}{25}$